

Atomic Design System Manual

Pangasinan Heritage Digital Showcase — Deliverable 1.2

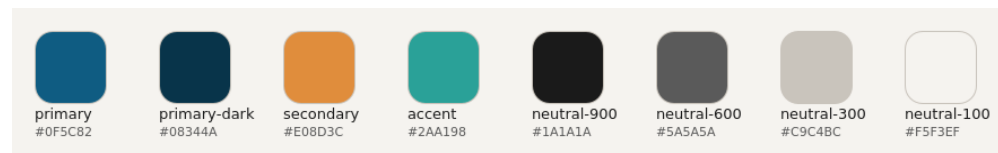
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This manual documents every component in the design system, organized by Brad Frost's Atomic Design methodology (atoms → molecules → organisms). Each entry includes a rendered visual preview, when and where to use the component, how it responds to different screen sizes, and a code reference showing how another developer can reuse it.

Design Tokens

All components draw their colors, spacing, and radii from a single token set, defined once in `app/globals.css` (CSS variables) and mirrored in `app/components/atoms/tokens.ts` for use in TypeScript. This keeps the palette consistent and makes a future re-theme a one-file change.



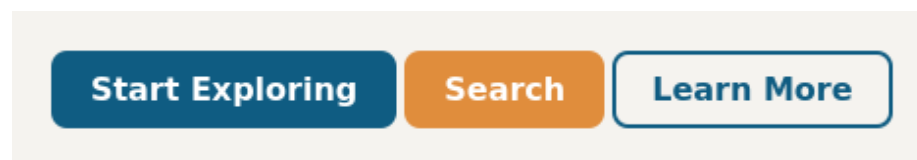
Color tokens: primary, primary-dark, secondary, accent, and the neutral scale used for text and backgrounds.

Atoms

Atoms are the smallest building blocks — they hold no business logic and are not aware of the Heritage Showcase content itself.

Atom — Button

1. Visual Preview



Primary, secondary, and outline variants at medium size.

2. Usage Context

Button is used for every clickable action in the app: the hero's “Start Exploring” call-to-action (primary), the SearchForm's submit action (primary), and any secondary/tertiary action such as “Learn More” (outline). Variant choice communicates action priority; it is never chosen for decoration alone.

3. Responsive Logic

- All screen sizes: padding and font-size stay fixed; only the surrounding layout wraps.
- Every size (sm/md/lg) keeps a minimum 44px height so touch targets remain accessible on mobile.

4. Code Reference

```
<Button variant="primary" size="lg" onClick={handleClick}>
  Start Exploring
</Button>

<Button variant="outline" size="sm" disabled>
```

[Learn More](#)
</Button>

Atom — Typography

1. Visual Preview

**H1 — Discover
Pangasinan**

H2 — Heritage Sites

H3 — Section Title

H4 — Card Title

Body — From the hundred limestone islands of Alaminos to the historic lighthouse of Bolinao.

Caption — Photo: Provincial Tourism Office

h1 through h4, body text, and caption variants.

2. Usage Context

Typography is used for every piece of text in the app instead of raw `<h1>/<p>` tags with inline classes, so heading scale and line-height stay consistent between the homepage hero, card titles, and footer captions.

3. Responsive Logic

- Mobile: h1/h2 use the smaller end of their responsive clamp (`text-4xl / text-3xl`).
- Tablet and up (md breakpoint): h1/h2 step up to `text-5xl / text-4xl` for more visual hierarchy on larger viewports.

4. Code Reference

```
<Typography variant="h1">  
  Discover Pangasinan's Living Heritage  
</Typography>
```

```
<Typography variant="caption">  
  Photo: Provincial Tourism Office  
</Typography>
```

Atom — Icon

1. Visual Preview



search, menu, close, location, and arrow-right icons.

2. Usage Context

Icon supplies every small pictogram in the app: the search field's magnifying glass, the mobile nav's hamburger/close toggle, the HeritageCard's location pin, and the “Explore →” link's arrow. It is decorative (aria-hidden) unless a title prop is supplied for icon-only buttons.

3. Responsive Logic

- Icons are fixed-size SVGs (size prop, default 24px) that do not themselves respond to breakpoints — the surrounding component decides icon size per context (e.g. 16px inside a card, 28px in the mobile nav toggle).

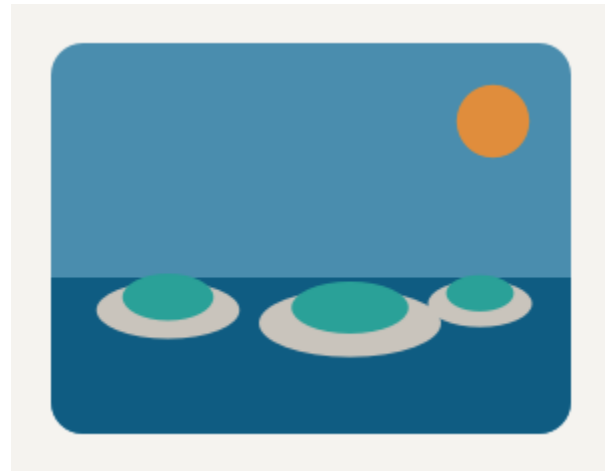
4. Code Reference

```
<Icon name="location" size={16} />

<button aria-label="Open menu">
  <Icon name="menu" size={28} title="Menu" />
</button>
```

Atom — Image

1. Visual Preview



A locked 4:3 aspect-ratio image, used as the base for HeritageCard's photo.

2. Usage Context

Image wraps next/image and is used anywhere a photo or illustration appears — currently the HeritageCard photo. It enforces a required alt prop and a locked aspect ratio so the layout never shifts while the image streams in on slow connections.

3. Responsive Logic

- All screen sizes: the aspect ratio (16/9, 4/3, or 1/1) is fixed; only the container's width changes, so height scales proportionally at every breakpoint.
- The sizes attribute tells the browser to request a full-width image on mobile and a third-width image on desktop grids, saving mobile data.

4. Code Reference

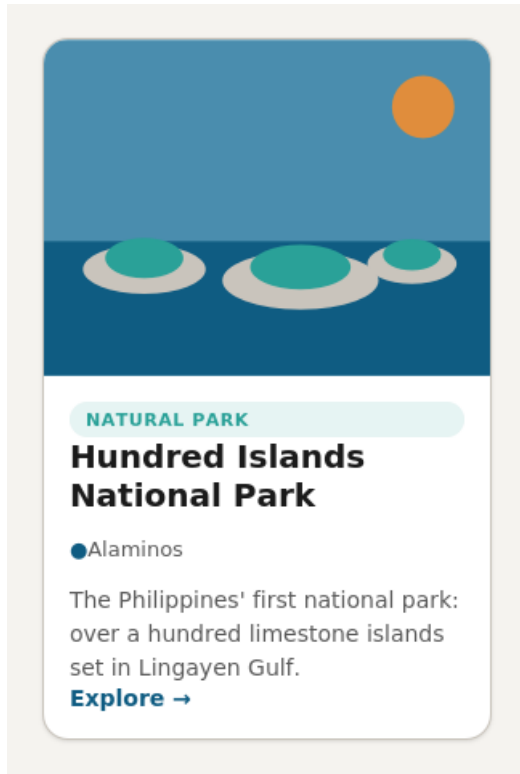
```
<Image
  src="/images/hundred-islands.svg"
  alt="Illustration of Hundred Islands"
  ratio="4/3"
/>
```

Molecules

Molecules combine a small number of atoms into a reusable, still-generic unit.

Molecule — Heritage Card

1. Visual Preview



A Heritage Card showing the Hundred Islands entry.

2. Usage Context

The Heritage Card is used exclusively for displaying tourist-site previews in the responsive heritage-site grid (HeritageGrid organism). It is not reused for unrelated list content — a different molecule should be created if another card-like list appears later.

3. Responsive Logic

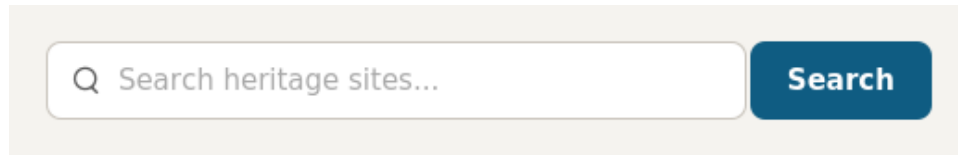
- Mobile (1-column grid): the card takes the full content width.
- Tablet (2-column grid, $\geq 640\text{px}$) and desktop (3-column grid, $\geq 1024\text{px}$): the card's width is set by its grid track, but its own internal layout (image → tag → title → meta → description → link) never changes — only the number of cards per row changes.

4. Code Reference

```
<HeritageCard
  slug="hundred-islands"
  title="Hundred Islands National Park"
  municipality="Alaminos"
  description="..."
  imageSrc="/images/hundred-islands.svg"
  tag="Natural Park"
/>
```

Molecule — Search Form

1. Visual Preview

A search input field with a magnifying glass icon and a blue 'Search' button. The input field has a placeholder text 'Search heritage sites...'. The button is blue with white text.

A search input with a magnifying-glass icon and a submit button.

2. Usage Context

The Search Form lets visitors filter the Heritage Grid by site name or municipality. It is kept separate from HeritageGrid so it can be reused on a future listing page (e.g. an events page) without duplicating markup.

3. Responsive Logic

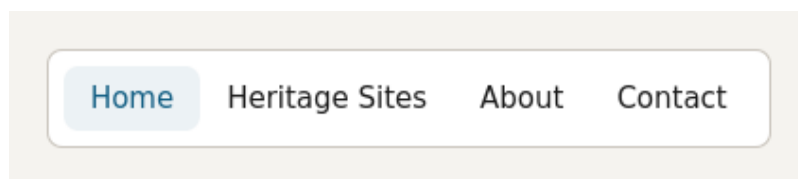
- Mobile: the input and button sit in one row; the input flexes to fill remaining width so the button never wraps to its own line.
- Desktop: the form's max-width caps at 480px so it does not stretch awkwardly across a wide hero section.

4. Code Reference

```
<SearchForm
  onSearch={(query) => setQuery(query)}
  placeholder="Search heritage sites..."
/>
```

Molecule — Navigation Item

1. Visual Preview



Four navigation items with the first one shown in its active state.

2. Usage Context

Navigation Item renders a single link inside HeaderNavigation, for both the desktop inline nav bar and the mobile slide-down panel — so hover/active states are defined once and stay visually identical in both contexts.

3. Responsive Logic

- Desktop: items sit in a horizontal row with small gaps.
- Mobile: the same items stack vertically at full width inside the slide-down panel, with slightly larger touch padding.

4. Code Reference

```
<NavigationItem
  href="/#sites"
  label="Heritage Sites"
  isActive={activePath === '/#sites'}
/>
```

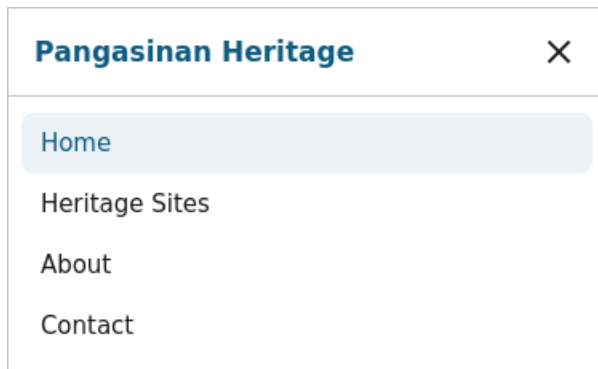
Organisms

Organisms assemble molecules (and atoms) into a distinct, page-level section.

Organism — Header Navigation

1. Visual Preview

Desktop: logo and nav items in one row.



Mobile: logo plus a hamburger toggle that reveals a slide-down panel.

2. Usage Context

Header Navigation appears once per page, in the root layout, above all page content. It is the site's primary wayfinding organism.

3. Responsive Logic

- < 768px (mobile): only the logo and a hamburger button are shown; tapping the button toggles a full-width slide-down panel (aria-expanded is kept in sync for screen readers).
- ≥ 768px (tablet/desktop): the hamburger button is hidden and the same nav items render inline next to the logo, so no click is needed to see the full menu.

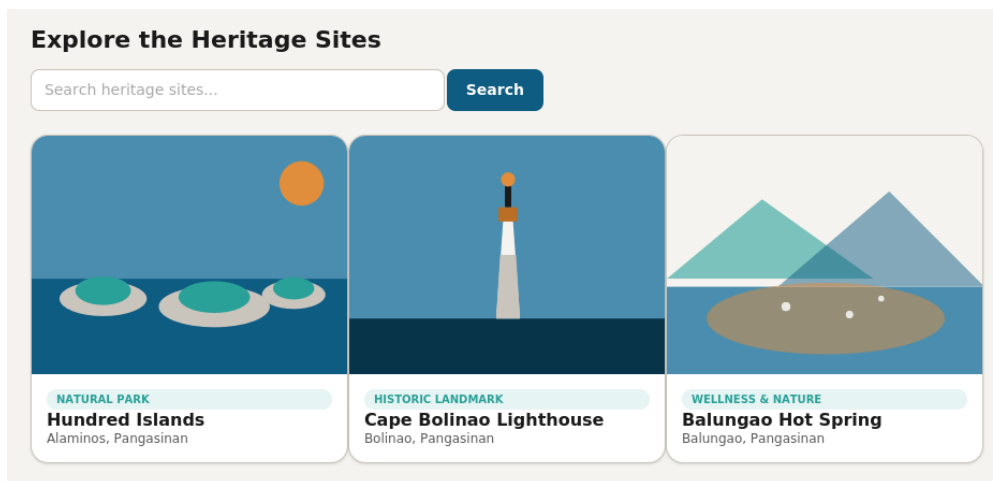
4. Code Reference

```
<HeaderNavigation activePath="/" />

// Internally:
// <nav className="hidden md:flex ..."> (desktop)
// {isOpen && <nav id="mobile-nav-panel">...} (mobile)
```

Organism — Heritage Grid

1. Visual Preview



Search form above a responsive grid of Heritage Cards (desktop, 3 columns).

2. Usage Context

Heritage Grid is the homepage's main content section: a Search Form above a filterable grid of Heritage Cards, one per tourist site (Hundred Islands, Cape Bolinao Lighthouse, Balungao Hot Spring).

3. Responsive Logic

- Mobile (< 640px): 1 column — cards stack full-width beneath the search bar.
- Tablet (≥ 640px): 2 columns.
- Desktop (≥ 1024px): 3 columns, as shown in the preview above.
- Filtering runs client-side against already-loaded data, so narrowing results never triggers a new network request — important on limited mobile data.

4. Code Reference

```
<HeritageGrid sites={heritageSites} />  
  
// heritageSites: HeritageSite[] from app/data/heritageSites.ts
```

Notes on Extending This System

- New atoms/molecules/organisms should live in the matching folder under app/components/ and, wherever possible, consume colors/spacing from tokens.ts rather than hard-coded values.
- Every new interactive element should keep the 44px minimum touch target and a visible focus-visible outline, consistent with the components documented above.
- This document should be updated whenever a component's props, responsive behavior, or visual design changes, so it continues to describe the actual implementation as required by the submission guidelines.